

Searching Algorithm:-

Linear Search:-

list = [12, 8, -2, 0, 1, 99, 11, 3, 4]
 0 1 2 3 4 5 6 7 8

target = 11

O/P → 6 | True or false

i ↓
 [12, 8, -2, 0, 1, 99, 11, 3, 4]
 0 1 2 3 4 5 6 7 8

O/P → True

i ↓ → out of Bound
 [12, 8, -2, 0, 1, 99, 11, 3, 4]
 0 1 2 3 4 5 6 7 8 9

target = 7

return → false

Algorithm:-

- 1) Initializing — $i = 0$
- 2) Using loop & search for target

```
for (i = 0 → len)
  if (arr[i] == target) {
    TRUE
  }
```

return false

Drawbacks of Linear Search:-

Size = 8 → less time
 assume = 10^{20}

Slower for large input

Binary search:-

[12, 8, -2, 0, 1, 99, 11, 3, 4]
0 1 2 3 4 5 6 7 8

Note: Array must be sorted

[12, 8, -2, 0, 1, 99, 11, 3, 4] $\xrightarrow{\text{Sorting}}$ [-2, 0, 1, 3, 4, 8, 11, 12, 99]
0 1 2 3 4 5 6 7 8 0 1 2 3 4 5 6 7 8

low $\downarrow$ high $\downarrow$

1). mid $\checkmark \rightarrow \frac{\text{low} + \text{high}}{2}$

Size = n

2) take variables / pointers $\rightarrow$ low $\rightarrow 0$
high $\rightarrow n-1$

3) observation & find $\rightarrow$ target

low $\downarrow$ mid $\downarrow$ low $\downarrow$ high $\downarrow$
[-2, 0, 1, 3, 4, 8, 11, 12, 99]
0 1 2 3 4 5 6 7 8

low $\downarrow$ high $\downarrow$ mid $\downarrow$ high $\downarrow$
[-2, 0, 1, 3, 4, 8, 11, 12, 99]
0 1 2 3 4 5 6 7 8

target = 11

low = 0

high = n-1

while (low $\leq$ high) :

mid = $\frac{\text{low} + \text{high}}{2}$

Case 1: Target is present mid pointer

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$\text{if}(\text{arr}[\text{mid}] == \text{target}) \rightarrow \text{TRUE}$

Case 2: Target is greater than mid $\rightarrow$ eliminate the left search space

$\text{low} = \text{mid} + 1$

Case 3: Target is lesser than mid

$\text{high} = \text{mid} - 1$

$\hookrightarrow$ Come out of the loop

return false

